Principle of Java Programming

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Java is a programming language used to create software, websites, mobile apps, and desktop applications

Java is:

- A **high-level programming language**
- **Easy to learn**
- **Object-oriented**
- **Platform independent**

Java programs are written once and can run on different systems.

Step 2: Install JDK

To write and run Java programs, install the Java Development Kit (JDK).

The JDK provides:

- `javac` → compiler
- `java` → program runner

Object-Oriented Programming (OOP).

Main concepts:

1. **Class** → blueprint
2. **Object** → real thing created from class
3. **Inheritance** → reusing code
4. **Polymorphism** → many forms
5. **Encapsulation** → data protection
6. **Abstraction** → hiding detail

Work1:

fn: hello.java:

```
public class hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

Work2:

```
class si {
    public static void main(String[] args) {

        short p = 1000, t = 2, r = 10;

        double si;

        si = (p * t * r) / 100;

        System.out.println("si=" + si);
    }
}
```

Work3: Variables and Data Types

```
public class StudentInfo {
    public static void main(String[] args) {
        String name = "Ram";
        int age = 20;


        System.out.println(name);
        System.out.println(age);
    }
}
```